

6. Analogue Input Voltages and Currents

Input	Resistance (Ohms)	Voltage VIN (Volts)		Nominal Current At VIN (mA)	Pin Number	
		Min	Max		High	Low
Front intermittent wiper dwell pot	0.0	0.0	0.2	-1.1	J2/19	J2/19
	4700	2.3	2.7	-0.5		
Cabin temperature sensor thermistor (High Series Only)	0.0	0.0	0.2	-1.1	J4/8	—
	4700	2.3	2.7	-0.5		

Table 2 - Analogue Input Voltage and Currents

6.1 Front Intermittent Wiper Dwell Pot

To test front intermittent wiper dwell pot, apply two different voltages across the analogue input and check the specified BEM memory location for the correct data. The steps in completing this test are as follows:

- Set accessories feed (J2/12) high
- Connect input (J2/19) directly to return (J2/8), applying 0V to the input.
- Using CCP diagnostic commands read address Analogue1_ub1 in RAM (refer to appendix for address). As per the engineering specification, the nominal A/D realised value for a short circuit (0V) input should be between \$00 and \$1F. The byte read must be between \$00 and \$10
- Connect a 4K7 resistor between input (J2/19) and return (J2/8), applying an input voltage between 2.3 and 2.7 volts.
- Using CCP diagnostic commands read address Analogue1_ub1 in RAM. The byte read must be between \$6E and \$82.

6.2 Cabin Temperature Sensor and Fan Supply (High Series Only)

To test the cabin temperature sensor, activate the sensor and read a byte from a RAM location. The steps necessary to perform this test are:

- Connect the CTS to the CTS fan supply (J4/7), CTS fan return (J4/11), CTS thermistor (J4/8) and thermistor return (J4/9) on the BEM.
- Set the ignition feed (J1/1) and accessories feed (J2/12) high.
- Read the byte from the Cabin_Temp_AD address (refer to appendix for address). The byte read must be between \$64 and \$8C (10 – 30 degrees Celsius).
- Cabin Sensor Fan supply is controlled by **PORT F bit 7** output with ignition on. Use "PTFDiagnosticFlag_ubg" and "PTFDiagnostic_ubg" to control the software.

6.3 Sunload Sensor (High Series Only)

To test the Sunload sensor, apply two different voltages to the input (J2/17), and check a byte in RAM for the correct data. Follow the steps:

- Set the ignition feed (J1/1) and accessories feed (J2/12) high.
- Connect the input (J2/17) directly to the return (J2/6) to apply an input of 0V.
- Read the byte from the Sunload Sensor Driver address (refer to appendix for address). If the byte read is not between \$00 and \$05, then record the test as a failure.
- Connect a 1K2 resistor between the input (J2/17) and return (J2/6).
- Read the byte from the Sunload Sensor Driver address. If the byte read is not between \$64 and \$96, then record the test as a failure.

V _{SUNLOAD} (Volts)	TCAN (Hex)	V _{SUNLOAD} (Volts)	TCAN (Hex)	V _{SUNLOAD} (Volts)	TCAN (Hex)	V _{SUNLOAD} (Volts)	TCAN (Hex)
5.1	FF	3.8	BD-C5	2.5	7C-81	1.2	3B-3D
5.0	FF	3.7	B8-C0	2.4	77-7C	1.1	36-38
4.9	FF	3.6	B3-BB	2.3	72-77	1.0	31-33
4.8	FF	3.5	AE-B5	2.2	6D-71	0.9	2C-2E
4.7	FF	3.4	A9-B0	2.1	68-6C	0.8	27-28
4.6	E5-EF	3.3	A4-AB	2.0	63-67	0.7	22-23
4.5	E0-EA	3.2	9F-A6	1.9	5E-62	0.6	1D-1E
4.4	DB-E4	3.1	9A-A0	1.8	59-5D	0.5	18-19
4.3	D6-DF	3.0	95-9B	1.7	54-57	0.4	13
4.2	D1-DA	2.9	90-96	1.6	4F-52	0.3	0E
4.1	CC-D5	2.8	8B-91	1.5	4A-4D	0.2	FF
4.0	C7-CF	2.7	86-8C	1.4	45-48	0.1	FF
3.9	C2-CA	2.6	81-86	1.3	40-42	0.0	FF

7. Digital Input Voltages

The following general procedure is used to test each digital input:

- Check all requirements are met before testing the digital input. For example some inputs require accessories feed high or a special connection.
- Apply a signal, based on characteristics in the Engineering Specification. To save time during final testing, ground inputs which require a voltage input low between 0 and x.xx volts.
- Read from the specified PTxInputDebounced_ubg address and check bit.
- Change the input state to the opposite state. For example if the input was grounded in the first state, apply 12V to the input.
- Read from the specified PTxInputDebounced_ubg address and check that the bit has changed.

Input	Method	Pass Criteria	Port & Bit	BEM Pin
Accessories Feed	1. Apply 12V to input 2. Ground input	Bit Set Bit Clear	Port G bit 0	J2/12
Ignition Feed	1. Apply 12V to input 2. Ground input	Bit Set Bit Clear	Port G bit 1	J1/1
Front Driver Door Input Switch	1. Ground input 2. Apply 12V to input	Bit Set Bit Clear	Port G bit 2	J1/14
Front Passenger Door Ajar Switch	1. Ground input 2. Apply 12V to input	Bit Set Bit Clear	Port T bit 1	J1/13
Rear Right Passenger Door Ajar Switch	1. Ground input 2. Apply 12V to input	Bit Set Bit Clear	Port T bit 0	J1/11
Rear Left Passenger Door Ajar Switch	1. Ground input 2. Apply 12V to input	Bit Set Bit Clear	Port T bit 3	J1/12
Boot/Tailgate Ajar Switch	1. Ground input 2. Apply 12V to input	Bit Set Bit Clear	Port H bit 0	J1/17
Driver Door Lock Status	1. Ground input 2. Apply 12V to input	Bit Set Bit Clear	Port T bit 4	J2/24
Front Intermittent Wiper Switch	1. Vacc high 2. Ground input 3. Apply 12V to input	Bit Set Bit Clear	Port C Bit 3	J1/19
Front Wiper Park Switch	1. Vacc high 2. Ground input 3. Apply 12V to input	Bit Clear Bit Set	Port D bit 7	J3/1
Front Wash Motor/Switch	1. Vacc high 2. Ground input 3. Apply 12V to input	Bit Set Bit Clear	Port B bit 2	J3/10
Washer Fluid Level Switch (High Series Only)	1. Vacc high 2. Ground input 3. Apply 12V to input	Bit Set Bit Clear	Port T bit 7	J1/20
Boot/Tailgate Release Switch	1. Ground input 2. Apply 12V to input	Bit Set Bit Clear	Port T bit 6	J1/9
Twilight Sensor (High Series Only)	1. Ground input 2. Apply 12V to input	Bit Set Bit Clear	Port F bit 0	J1/18
CAN global wake-up request	1. Ground input 2. Apply 12V to input	Bit Set Bit Clear	Port H bit 1	J2/21

8. Output Voltages

8.1 Relay Cleaning Process

It is recommended that Relay contact cleaning be performed using a 1A inductive load, cycled 5 times before making a measurement. The following relays are to be tested preferably using the ICT.

- Central Door Unlock Motor Relay (RL1)
- Central Driver Door Unlock Motor Relay (RL2)
- Central Door Lock Motor Relay (RL3)
- Boot/Tailgate Release Solenoid Relay (RL4)
- Battery Saver Supply Relay (RL5)
- Front Washer Motor Relay (RL6)
- Front Wiper Park Switch Relay (RL7)

8.2 Switched Output & Solid State Output Voltages

The general procedure for testing outputs is as follows:

- Set up any initial conditions before switching outputs. Connect loads to the BEM where applicable.
- Clear the specified bit of the PTxDiagnostic_ubg address.
- Set the specified bit of the PtxDiagnosticFlag _ubg to take control of the output.
- Measure the output & compare reading with the engineering specification.
- Set the specified bit of the PTxDiagnostic_ubg address to change the state of the output.
- Measure the output & compare reading with the engineering specification
- Clear the specified bit of the PtxDiagnosticFlag_ubg to release control of the output.

Output	Method	Pass Criteria	Port & Bit	BEM Pin
Battery Saver Supply	1. Connect load 2. Set diagnostic & flag buffer bits 3. Measure courtesy lamp fuse (V_s) and use as limit. 4. Measure battery saver supply & compare with output on limits. 5. Clear diagnostic buffer bit 6. Measure battery saver supply & compare with output off limits 7. Clear flag buffer bit.	<u>Output On</u> Min Max $V_s-0.5$ $V_s+0.1$ <u>Output Off</u> Min Max -0.15V 1.0V * Output is open circuit in off state	Port D Bit 4	J3/3
Front Wiper Motor	1. Set accessories feed high & connect wiper motor load. 2. Set diagnostic & flag buffer bits. 3. Measure J3/2 & compare with on limits. 4. Clear diagnostic buffer bit 5. Measure J3/2 & compare with off limits 6. Clear flag buffer bit.	<u>Output On</u> Min Max -0.15V 0.5V <u>Output Off</u> Min Max $V_{J3/1}$ $V_{J3/1}+0.5$	Port D Bit 6	J3/2
Central Door Unlock Motor	1. Connect door lock motor loads 2. Set diagnostic & flag buffer bits. 3. Measure central door lock fuse (V_s) 4. Measure J5/1, subtract from V_s & compare with output on limit. 5. Clear diagnostic buffer bit 6. Measure J5/1 and compare with output off limit. 7. Clear flag buffer bit.	<u>Output On</u> Min Max $V_s-0.3$ V_s <u>Output Off</u> Min Max -0.1V 0.4V	Port D Bit 0	J5/1

Output	Method	Pass Criteria	Port & Bit	BEM Pin
Driver Lock Motor	1. Connect door lock motor loads 2. Set diagnostic & flag buffer bits. 3. Measure central door lock fuse (V_s) 4. Measure J5/3, subtract from V_s & compare with output on limit. 5. Clear diagnostic buffer bit 6. Measure J5/3 and compare with output off limit. 7. Clear flag buffer bit.	<u>Output On</u> Min Max $V_s - 0.3$ V_s <u>Output Off</u> Min Max -0.1V 0.4V	Port D Bit 2	J5/2
Driver Door Unlock Motor	1. Connect door lock motor loads 2. Set diagnostic & flag buffer bits. 3. Measure central door lock fuse (V_s) 4. Measure J3/9, subtract from V_s & compare with output on limit. 5. Clear diagnostic buffer bit 6. Measure J3/9 and compare with output off limit. 7. Clear flag buffer bit.	<u>Output On</u> Min Max $V_s - 0.3$ V_s <u>Output Off</u> Min Max -0.1V 0.3V	Port D Bit 1	J3/9
Boot Tailgate Release Solenoid	1. Connect relay load 2. Set diagnostic & flag buffer bits. 3. Measure J3/7 & compare with on limit. 4. Clear diagnostic buffer bit 5. Measure J3/7 and compare with off limit. 6. Clear flag buffer bit.	<u>Output On</u> Min Max $V_s - 0.3V$ $V_s + 0.1$ <u>Output Off</u> Min Max -0.15V 0.3V	Port D Bit 3	J3/7
Front Wash Motor Switch (High Series Only)	1. Set accessories feed high & connect wash motor 2. Set diagnostic & flag buffer bits. 3. Measure J3/10 & compare with on limit. 4. Clear diagnostic buffer bit 5. Measure J3/10 and compare with off limit. 6. Clear flag buffer bit.	<u>Output On</u> Min Max -0.1V 0.5V <u>Output Off</u> Min Max $V_s - 1.0$ $V_s + 0.1$ *Output is open circuit in off state	Port D Bit 5	J3/10
Heated Backlight Relay	1. Set accessories feed high & connect relay load. 2. Set diagnostic & flag buffer bits. 3. Measure J2/10 & compare with on limit. 4. Clear diagnostic buffer bit 5. Measure J2/10 and compare with off limit. 6. Clear flag buffer bit.	<u>Output On</u> Min Max -0.1V 2.0V <u>Output Off</u> Min Max $V_s - 1.0$ $V_s + 0.1$	Port L Bit 7	J2/10
Horn Relay	1. Connect relay load. 2. Set diagnostic & flag buffer bits. 3. Measure J2/9 & compare with on limit. 4. Clear diagnostic buffer bit 5. Measure J2/9 and compare with off limit. 6. Clear flag buffer bit.	<u>Output On</u> Min Max -0.15V 2.0V <u>Output Off</u> Min Max $V_s - 1.0V$ $V_s + 0.1$	Port F Bit 2	J2/9
Power Window Relay	1. Connect relay load. 2. Set diagnostic & flag buffer bits. 3. Measure J2/11 & compare with on limit. 4. Clear diagnostic buffer bit 5. Measure J2/11 and compare with off limit. 6. Clear flag buffer bit.	<u>Output On</u> Min Max -0.15V 2.0V <u>Output Off</u> Min Max $V_s - 1.0V$ $V_s + 0.1$	Port L Bit 6	J2/11
Courtesy Lamp	1. Connect globe loads. 2. Turn on battery saver supply 3. Set diagnostic & flag buffer bits. 4. Measure J3/12 & compare with on limit. 5. Clear diagnostic buffer bit 6. Measure J3/12 and compare with off limit. 7. Clear flag buffer bit.	<u>Output On</u> Min Max -0.15V 1.5V <u>Output Off</u> Min Max $V_s - 1.0V$ $V_s + 0.1$	Port E Bit 3	J3/12

Output	Method	Pass Criteria	Port & Bit	BEM Pin
Memory Seat Enable (High Series Only)	1. Connect load. 2. Set diagnostic & flag buffer bits. 3. Measure J2/23 & compare with on limit. 4. Clear diagnostic buffer bit 5. Measure J2/23 & compare with off limit. 6. Clear flag buffer bit.	<u>Output On</u> Min Max -0.15V 2.0V <u>Output Off</u> Min Max $V_S-1.0V$ $V_S+0.1$	Port L Bit 4	J2/23
Memory Seat Data (High Series Only)	1. Connect load. 2. Set diagnostic & flag buffer bits. 3. Measure J2/22 & compare with on limit. 4. Clear diagnostic buffer bit 5. Measure J2/22 & compare with off limit. 6. Clear flag buffer bit.	<u>Output On</u> Min Max -0.15V 2.0V <u>Output Off</u> Min Max $V_S-1.0V$ $V_S+0.1$	Port L Bit 5	J2/22
Auto Headlamp Relay (A) (High Series Only)	1. Set accessories feed high & connect relay load. 2. Set diagnostic & flag buffer bits. 3. Clear Port A Bit 6 4. Measure J1/5 & compare with on limit. 5. Clear diagnostic buffer bit 6. Measure J1/5 & compare with off limit. 7. Clear flag buffer bit.	<u>Output On</u> Min Max $V_S-1.0V$ $V_S+0.1$ <u>Output Off</u> Min Max -0.15V 1.5V	Port F Bit 1	J1/6
Auto Headlamp Relay (B) (High Series Only)	1. Set accessories feed high & connect relay load. 2. Set diagnostic & flag buffer bits. 3. Measure J2/1 & compare with on limit. 4. Clear diagnostic buffer bit 5. Measure J2/1 & compare with off limit. 6. Clear flag buffer bit.	<u>Output On</u> Min Max -0.15V 1.5V <u>Output Off</u> Min Max $V_S-1.0V$ $V_S+0.1$	Port A Bit 6	J2/1
Transceiver Supply Measure	1. Set diagnostic & flag buffer bits. 2. Measure J1/7 & compare with on limit. 3. Clear diagnostic buffer bit 4. Measure J1/7 & compare with off limit. 5. Clear flag buffer bit.	<u>Output On</u> Min Max -0.15V 1.0V <u>Output Off</u> Min Max $V_S-2.0V$ $V_S+0.1$	Port C Bit 1	J1/4
CTS Fan Supply (High Series Only)	1. Set diagnostic & flag buffer bits. 2. Measure J4/7 & compare with on limit. 3. Clear diagnostic buffer bit 4. Measure J4/7 & compare with off limit. 5. Clear flag buffer bit.	<u>Output On</u> Min Max $V_S-1.0V$ $V_S+0.1$ <u>Output Off</u> Min Max -0.15V 0.5V	Port F Bit 6	J4/7
CAN Global Wakeup request	1. Set diagnostic & flag buffer bits. 2. Measure J2/21 & compare with on limit. 3. Clear diagnostic buffer bit 4. Measure J2/21 & compare with off limit. 5. Clear flag buffer bit.	<u>Output On</u> Min Max -0.15V 2.0V Min Max $V_S-1.0V$ $V_S+0.1$	Port L Bit 0	J2/21

9. Watchdog / Reset Tests

9.1 Mutual Watchdog – Temic Test

Steps required for the Temic microprocessor to be reset by the Motorola microprocessor are:

- Clear f_Temic_reset_g address (bit 1)
- Set BEM_MainFlags_ubl address (bit 6 of lower byte)
- Set CrashKey_uwl address to 55AA
- Wait up to 2.5 seconds for the Temic microprocessor to reset
- Re-establish CCP communication. If CCP cannot be re-established, record failure.
- Read f_Temic_reset_g address bit 1. If this bit has changed back to 1, then the Temic microprocessor has been successfully reset.

9.2 Mutual Watchdog – Motorola Test

Steps required to reset the Motorola microprocessor are:

- Set BEM_MainFlags_ubl.HC08CrashTest address bit 5.
- Set CrashKey_uwl address to 55AA.
- Wait up to 1.5 seconds for the Motorola microprocessor to reset.
- Re-establish CCP communication. If CCP cannot be re-established, record failure.

10. Quiescent Current

The quiescent current is measured when the BEM is in sleep mode, with all inputs and outputs at levels described in the engineering specification. The steps required to measure the quiescent current are shown below:

- Connect the current shunt, then disconnect main battery fuse, courtesy lamp fuse and central door locking fuse. This will allow the current being drawn by the battery fuse, courtesy lamp fuse and central door-locking fuse to be measured across a shunt resistor.
- Turn supply on, wait 500ms before enabling CCP.
- Set accessories and ignition feed low and ensure no digital inputs or switched outputs are in an active state, except the RF receiver supply.
- Get the BEM into sleep (Stop) mode by setting the following:
 - Set RAM CrashKey_uwl at address \$C8 and \$C9 to \$55 and \$AA,
 - Set Bit 4 of RAM BEMMainFlags_uhl at address \$52 to 1
 - BEM will go into STOP mode within 200ms
- Perform voltage compensation, to ensure that the quiescent current measurement is measured with a supply voltage of 12V.
- Setup the digital multimeter to sample the voltage across the shunt. Find the voltage samples average and divide by the resistance of the shunt to obtain the quiescent current measurement.
- Connect the main battery fuse, courtesy lamp fuse and central locking fuse and disconnect the current shunt. Set accessories feed to high to exit sleep mode.

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11. Low Voltage Reset

Use the heated backlight relay output to perform this test. Whilst monitoring the heated backlight output voltage, set the supply voltage to 6V, then 4.5V. For a successful low voltage reset, the output in its on state should change from 0V to 4.5V, indicating the BEM has reset due to the low supply voltage. Steps to perform this test are:

- Connect the heated backlight relay load and set accessories feed high.
- Turn on the heated backlight relay by setting the diagnostic and flag buffer bits (Port L, Bit 7).
- Set the power supply voltage to 6V and measure the output voltage. Check the output voltage is between -0.15 and 2.0V.
- Set the power supply to 4.5V and measure the output voltage. Check that the BEM has successfully reset by checking the output voltage is between 3.5 and 4.6V.

12. Software Version & BEM Variant

12.1 Software Version Number

Read the software version number using the following Ford diagnostic command.

Message	Protocol	Transmit	Receive
Software Ver Number	FD	03 22 E2 00 00 00 00 00	06 62 E2 00 ab cc dd 00 (Response with data)
			03 7F 22 78 00 00 00 00 (Busy, repeat request again)

Return data format:

- a = software version number (0-15) encoded as hexadecimal
- b = software date (month) (0-12) encoded as hexadecimal
- cc = software date (day) (0-31) encoded as hexadecimal
- dd = software date (year) (0-255) encoded as hexadecimal with an offset of 1900 (65 is 2001)

Pass:V13, dated 6/8/2002, other fail test

12.2 BEM Variant

The BEM variant can be verified by reading address BEMMainFlags_ubl.HardwareDetected. If the bit is set, the BEM is a high series, otherwise the BEM is a low series.

For P&A, seven EEprom locations are to be modified (See Sect 15) at addresses \$800-\$803, \$840, \$870 and \$8EB. The auto EEprom download is applicable only for OE, therefore the Final Tester or ICT will have to write and verify the specific P&A data.

BEMMainFlags_ubl [2]	FD4A	First byte: PITHappened Bit 0: //Triggers EEPROM Manage outside PIT Interrupt PowerUp Bit 1: //Indicates that one 100ms loop has passed since // power up for the purpose of assuring that // modules can power up cleanly. DiagnosticMode Bit 2: //Indicates when diagnostic mode is active HardwareDetected Bit 3: //Records hardware detected at startup -1 = high series StopModeTest Bit 4: //Test quiescent current in stop mode HC08CrashTest Bit 5: //Test that HC08 crash is recovered by Temic TemicCrashTest Bit 6: //Test that Temic crash actually resets Temic TemicShutdownRequested Bit 7: //Want Temic interface in shutdown for STOP mode Second Byte: TemicShutdownReceived Bit 0: //Temic interface in shutdown mode Temic_Error Bit 1: //Indicates Temic error - Temic reset in progress Temic_Reset_Initiated Bit 2: ECU_Programming_Mode Bit 3: ECU_Adjustment_Mode Bit 4: uchar EEPROM_Load Bit 5:
f_flag_spi_g	FD4C	Set bit 1 to test the function of Temic requesting Motorola to reset.
ADwiperpot_ubg	FD4E	A to D reading from wiper potentiometer.
Cabin_Temp_AD_reading_ubg	FD50	A to D reading of cabin temperature sensor
SunloadSensorAD_ubg	FD52	A to D reading of sunload sensor
a_mgbuff_ubg[4]	FD54	Reports remote key pressed. (Lock key press gives \$10)

17.4 CCP Encryption Algorithm

The following is the code used in Barra BEM for calculating values to be used in CCP Unlock command.

```
#define SECURITY_ID0    0x41
#define SECURITY_ID1    0x55
#define SECURITY_ID2    0x36
#define SECURITY_ID3    0x74

unsigned char challenge0_ubl, challenge1_ubl, challenge2_ubl, challenge3_ubl;
unsigned char result0_ubl, result1_ubl, result2_ubl, result3_ubl, reply_cnt_ubl;

void Calculate_Result(void)
{
    unsigned int  adden_a_uil, adden_b_uil, sum_uil;
    unsigned char temp1_ubl, temp2_ubl, temp3_ubl;

    adden_a_uil = (unsigned int) (SECURITY_ID2 << 8) + (unsigned int) SECURITY_ID3;
    adden_b_uil = (unsigned int) (challenge2_ubl << 8) + (unsigned int) challenge3_ubl;
    sum_uil = adden_a_uil + adden_b_uil;

    result2_ubl = (unsigned char) (sum_uil >> 8);
    result3_ubl = (unsigned char) (sum_uil & 0xff);

    adden_a_uil = (unsigned int) (SECURITY_ID0 << 8) + (unsigned int) SECURITY_ID1;
    adden_b_uil = (unsigned int) (challenge0_ubl << 8) + (unsigned int) challenge1_ubl;
    sum_uil = adden_a_uil + adden_b_uil;

    result0_ubl = (unsigned char) (sum_uil >> 8);
    result1_ubl = (unsigned char) (sum_uil & 0xff);

    result3_ubl = result3_ubl * 13;
    result2_ubl = result2_ubl * 11;
    result1_ubl = result1_ubl * 19;
    result0_ubl = result0_ubl * 17;

    result3_ubl = (((temp1_ubl = result3_ubl) << 6) | ((temp2_ubl = result3_ubl) >> 2));
    result2_ubl = (((temp1_ubl = result2_ubl) << 6) | ((temp2_ubl = result2_ubl) >> 2));
    result1_ubl = (((temp1_ubl = result1_ubl) << 6) | ((temp2_ubl = result1_ubl) >> 2));
    result0_ubl = (((temp1_ubl = result0_ubl) << 6) | ((temp2_ubl = result0_ubl) >> 2));

    result3_ubl = result3_ubl ^ 17;
    result2_ubl = result2_ubl ^ 86;
    result1_ubl = result1_ubl ^ 75;
    result0_ubl = result0_ubl ^ 52;

    if (result2_ubl & 0x04)
    {
        temp3_ubl = (temp1_ubl = (result3_ubl >> 4)) | (temp2_ubl = (result3_ubl << 4));
        temp2_ubl = (temp1_ubl = (result2_ubl >> 4)) | (temp2_ubl = (result2_ubl << 4));
        result3_ubl = temp2_ubl;
        result2_ubl = temp3_ubl;
    }
    else
    {
        temp3_ubl = (temp1_ubl = (result1_ubl >> 4)) | (temp2_ubl = (result1_ubl << 4));
        temp2_ubl = (temp1_ubl = (result0_ubl >> 4)) | (temp2_ubl = (result0_ubl << 4));
        result1_ubl = temp2_ubl;
        result0_ubl = temp3_ubl;
    }
}
```

17.5 Calculate DST Encryption Key

The following procedure is used to convert the serial number into the DST Encryption Key

1. Convert the serial number to 4-byte hexadecimal form. Perform rotates on the 4 SN bytes as follows (rotate right, putting least significant bit into most significant bit):
2. SN[0]: do not rotate
3. SN[1]: rotate right once
4. SN[2]: rotate right 5 times
5. SN[3]: rotate right 3 times

Calculate bytes as follows:

1. Byte[0] = 02 (EBEM2 was 00, GBEM 01)
2. $\text{Byte}[1] = ((\text{SN}[1] \& 0F) \ll 4) + (\text{SN}[3] \& 0F) \wedge 3A$
3. $\text{Byte}[2] = (\text{SN}[2] \& F0) + ((\text{SN}[0] \& F0) \gg 4) \wedge D0$
4. $\text{Byte}[3] = ((\text{SN}[3] \& 0F) \ll 4) + (\text{SN}[0] \& 0F) \wedge 53$
5. $\text{Byte}[4] = (\text{SN}[2] \& F0) + ((\text{SN}[3] \& F0) \gg 4) \wedge 2C$

